

Career XRcade Framework: Student-driven Collaboration Processes to Develop Learning Environments for Immersive Career Exploration

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ABSTRACT

The Career XRcade (CXR) Framework is our platform to allow college students creating educational immersive learning experiences that prepare high school students to explore diverse career paths. These virtual journeys through various career paths offer students a medium to understand real-world job environments and demands. We have demonstrated the use of our framework through the creation of two virtual worlds each showcasing 5 career paths in Cybersecurity and Esports industries. Our proposed user studies on our case studies will gather data and insights from our stakeholders, aiming to measure the effectiveness of such a framework in academic settings.

Index Terms: Immersive Learning—Extended Reality—Collaborative and co-creative design frameworks—Career identity

1 INTRODUCTION

While the application of immersive experiences can help train students to consider future career paths, there is a lack of support for designing and prototyping such experiences, especially for complex Extended Reality(XR) environments [5]. Development of these applications is traditionally expensive and unscalable to the multitude of emerging careers. We aim to study collaborative patterns for efficiently building immersive experiences through our Career XRcade (CXR) Framework, a platform for college students to collaboratively create educational XR (Extended Reality) experiences that prepare high school students to explore diverse career paths through interactive, explorable exhibits. These virtual journeys through various career paths offer students an innovative medium to understand real-world job environments and demands. Rooted in co-creative and cross-functional collaboration with subject matter experts, immersive platforming for student engagement, and student-led design for scalable development, the CXR process aims to streamline educational XR application design and development, making it scalable, accessible, and adaptable within the educational milieu.

The produced CXR Lands will engage students through explorable rooms of interactive exhibits, including mini-games, puzzles, informational panels, and fun gizmos that showcase specific career paths. Each Career Land includes a Hub and five Career Exhibits (Figure 1). Using desktops or VR headsets, students can explore the Lands, jumping into the various visually appealing gamified experiences and embarking on a story-driven dynamic learning journey, discovering future careers to thrive in fast-growing industries such as Cybersecurity and Esports. Developed through our collaborative framework, these Lands align with a structured career curriculum. Career XRcade Lands aims to provide a sense of presence and exploration for learners to positively impact their motivation, self-efficacy, and learning outcomes [1].

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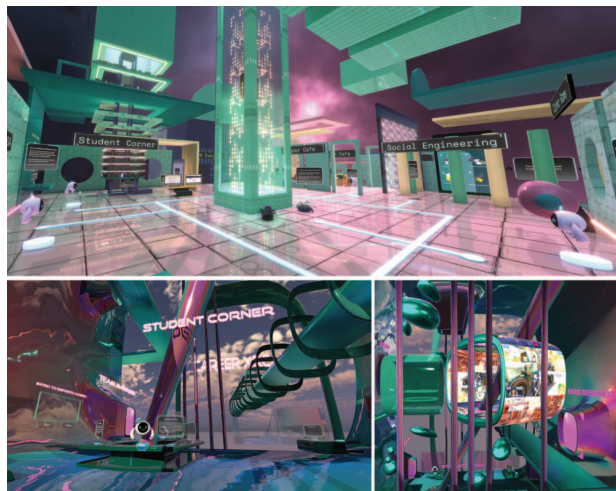


Figure 1: Immersive Career Lands - Cybersecurity Land and Esports Land Hubs

2 COLLABORATIVE FRAMEWORK DEVELOPMENT PROCESS

The CXR Collaborative Framework streamlines development through student-led design sessions, accessible design templates/toolkits, and iterative expectations. The framework consists of (i) project roadmapping, (ii) involvement of subject matter experts (instructional designers and industry professionals), (iii) integration of curriculum and learning objectives, and (iv) collaborative XR design and development. This iterative process targets continuous improvement and alignment with industry needs, towards an effective learning experience for students [7].

Project Roadmapping: The initial phase establishes project goals, objectives, and roadmaps. This process uses the CXR templates as an initial blueprint for collaborative efforts between students and professionals to achieve career-oriented goals.

Connecting Industry Expert Stakeholders: The next phase engages industry experts to gain insights into real-world job demands and career trajectories. This step serves a dual purpose: to secure industry sponsorship and funding for the development and to ensure the developed immersive experience reflects various career paths. In our case studies, the student developers connected with a telecommunications company. Through a series of meetings, the team collaboratively examined the proposed roadmaps and desired curriculum. This step is instrumental in aligning the immersive experiences with industry expectations, thus enriching the content with practical and industry-relevant perspectives.

Curricular Integration with Learning Objectives: Through curriculum integration – especially defining the learning objectives of each immersive experience – the framework aligns with educational content and the needs of its target audience. Students led ideation workshops with instructional designers. In these workshops participants used CXR Ideation and User Journey templates in Miro to brainstorm ideas that assist them in visualizing specific career paths.



Figure 2: Career Exhibits: Esports Team Manager, Esports Game Designer, Cybersecurity Blue Team and Cybersecurity SOC Analyst

Immersive Experiences Development and Feedback Loop: The development of immersive learning experiences within the Career XRcade framework engages student artists, designers, and developers in a comprehensive design and development process. Considering the literature on design methods in human-centered design for AR and VR applications such as [6] and [2], the process is facilitated by a range of accessible design and development resources, including design templates, worksheets, ideation and user experience workshops, and 3D assets. The students assemble these elements through the Unity Engine, designing the level, programming the Desktop and VR interactions, and building and testing for desktop (WebGL version) and VR headsets.

The implementation of the CXR framework entails an iterative process involving meetings, pilot studies, focus groups, and ongoing assessments. These allow for the refinement of the framework based on insights and data gathered from different groups of stakeholders involved in the process. This iterative approach ensures the framework's continual evolution and adaptation, aligning it with the evolving needs of job demands and instructional curriculum.

3 EARLY CASE STUDIES

Our initial application of the Career XRcade framework resulted in a Cybersecurity Land and an Esports Land. These lands were created for Desktop and VR through a multidisciplinary team of student developers. These immersive learning environments incorporate elements of roleplaying, problem-solving, virtual assistants, reward systems, and gamified assessments to ensure student engagement.

Cybersecurity Land includes interactive experiences providing general cybersecurity information, a student corner with an interactive library, the social engineering exhibit, along with a Cyber cafe. The Hub also serves as a gateway leads to five career exhibits: Information Security Analyst, Security Operation analyst, Blue Team Analyst, Red Team Analyst, and Mobile Architect. A robot-like virtual assistant, Holobuddy, guides the user throughout the land. Each career exhibit consists of mini-games, interactive vocabulary activities, gadgets, puzzles, quizzes, informational panels, interviews, and fun gizmos that illuminate various facets of the respective careers. A Gem Collectable system tracks user progress through these exhibits.

Esports Lands, our second iteration of using the framework's application, echoes similar features with a vibrant aesthetic considering the Esports career spirit. Its hub has a jumbotron animated screen displaying popular games along with portals and elevators leading to five Career Exhibits: Game Designer, Game Developer, Team Manager, Esports Marketer, and Content creator (Figure 2). Each Career Exhibit employs distinct narrative and gamification strategies to visualize career paths and expose students to the job concepts, responsibilities, and values of an expert in these industries.

4 PLANNED USER STUDIES

We planned a quasi-experimental study to test the effectiveness of the Career XRcade Framework with different stakeholders: 60 high school students, 3 teachers, 15 college student developers, 2 instructional designers, and 2 career experts. Our research involves a series of workshops, surveys, and interview questions for each group involved in our case studies.

Student Users: A pre-survey and post-survey have been designed for high school students, assessing their career interests after using the Career XRcade VR application. The survey design is inspired and adapted from "VRUSE" [3] and "STEM-CIS" [4], investigating the usability of the VR app and developing career interest in cybersecurity and Esports careers as our case studies. We hypothesize that our immersive exhibits facilitate a deeper exploration of those career paths, aiding in the development of a more concrete career identity at an early stage.

High School Instructional Designers and Teachers: We conducted an in-person workshop with teachers, instructional designers, and student developers at a high school in Texas. This workshop focused on the use of VR technology and on interacting within the Career XRcade VR app; teachers explored career exhibits in the virtual world. A focus group and interview questions explore Career XRcade app usability, curricular relevance, and potential impact on student's career identity to explore teachers' experience with the app and self-assessed expertise.

Student Developers: Our student developers have backgrounds in art, design, computer science, and engineering. Through the framework's student-driven spirit, the team has provided insights throughout the design and development process. We designed a Likert Scale survey and a series of open-ended questions to investigate student perspectives further. Applying co-creative participatory design, agile methodologies, and design thinking processes, we will explore how these methods build necessary skills/competencies. Also, we will investigate how their contribution to this project affects their self-efficacy, career identity, and aspirations.

Instructional Designers and Industry Professionals: Our case studies have engaged two instructional designers and two subject matter experts. Through studying their engagement, we will evaluate the framework's curricular alignment and its relevance to real-world job demands.

REFERENCES

- [1] A. Dengel. What is immersive learning? In *2022 8th International Conference of the Immersive Learning Research Network (iLRN)*, pp. 1–5. IEEE, 2022.
- [2] C. Hillmann. *UX and Experience Design: From Screen to 3D Space*, pp. 117–155. Apress, Berkeley, CA, 2021. doi: 10.1007/978-1-4842-7020-2_4
- [3] R. S. Kalawsky. Vruse—a computerised diagnostic tool: for usability evaluation of virtual/synthetic environment systems. *Applied ergonomics*, 30(1):11–25, 1999.
- [4] M. W. Kier, M. R. Blanchard, J. W. Osborne, and J. L. Albert. The development of the stem career interest survey (stem-cis). *Research in Science Education*, 44:461–481, 2014.
- [5] V. Krauß, M. Nebeling, F. Jasche, and A. Boden. Elements of xr prototyping: Characterizing the role and use of prototypes in augmented and virtual reality design. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, CHI '22. Association for Computing Machinery, New York, NY, USA, 2022. doi: 10.1145/3491102.3517714
- [6] R. C. Stevens. *Designing Immersive 3D Experiences: A Designer's Guide to Creating Realistic 3D Experiences for Extended Reality*. New Riders, 2021.
- [7] A. Varella, P. E. Antoniou, J. D. Pickering, C. Chatzimallis, and P. D. Bamidis. Evaluating co-creative xr resource design and development; observations from the field. In *2022 8th International Conference of the Immersive Learning Research Network (iLRN)*, pp. 1–5. IEEE, 2022.